Regression model

$$d(p_1, p_2) = \beta_0 + \beta_1 p_1 + \beta_2 p_2$$

observed values $\hat{d}$ are integers

Typical objective function(s)

$$\sum_{i=1}^{n} \left(\hat{d}_i - d(p_{1i}, p_{2i})\right)^2 \qquad \text{OLS}$$

or $$\frac{1}{n} \sum_{i=1}^{n} \left|\hat{d}_i - d(p_{1i}, p_{2i})\right| \qquad \text{LAD}$$

We have incorporate the time t,
integers values, and
slightly differing parameters w.r.t. t

Assumptions

- d is monotonously decreasing in p_1
- d is monotonously increasing in p_2
- d is slightly depending on time,
  i.e. d is monotonously increasing in time $t = i$
  $\Rightarrow d(p_1, p_2, t) = \beta_{0t} + \beta_{1t} p_1 + \beta_{2t} p_2$
- d and its parameters $\beta_{0\cdot}, \beta_{1\cdot}, \beta_{2\cdot}$ does not differ to much for t and $t-1$, $t+1$

$$|\beta_{kt} - \beta_{kt-1}| \leq \delta$$

- d is symmetric in p_1 and p_2

my demand d_{own}
comp. demand d_{comp}

$$\left.\begin{aligned} d(p_1, p_2, t) &= d_{own} \\ d(p_2, p_1, t) &= d_{comp} \end{aligned}\right\} \text{"fair split" of demand}$$

Linearization of the ceil function and the floor function

floor: $y = \lfloor x \rfloor$ x is rounded down to the next integer

examples: $x = 0.001$
$x = 0.9$
$x = 0.5$
$x = 1.1$

additional integer variable $y \in \mathbb{N}_0$

$$x - 0.999 \leq y \leq x \quad \left.\right\} \text{ 2 constraints } \quad \begin{aligned} x - 0.999 &\leq y \\ y &\leq x \end{aligned}$$

examples:
$0.001 - 0.999 = -0.998 \leq y \leq 0.001 \Rightarrow y = 0$
$0.9 - 0.999 = -0.009 \leq y \leq 0.9 \Rightarrow y = 0$
$0.5 - 0.999 = -0.499 \leq y \leq 0.5 \Rightarrow y = 0$
$1.1 - 0.999 = 0.101 \leq y \leq 1.1 \Rightarrow y = 1$

ceil $y = \lceil x \rceil$ round up to the next integer

$y \in \mathbb{N}_0$: $x \leq y \leq x + 0.999$ $\}$ 2 constraints $\quad x \leq y$, $\quad y \leq x + 0.999$

examples: $x = 0.5 \Rightarrow 0.5 \leq y \leq 0.5 + 0.999 = 1.499 \Rightarrow y = 1$
$x = 1.1 \Rightarrow 1.1 \leq y \leq 1.1 + 0.999 \leq 2.099 \Rightarrow y = 2$

rounding to the next integer:

$y \in \mathbb{N}_0$: $x - 0.499 \leq y \leq x + 0.5$ $\}$ 2 constraints $\quad x - 0.499 \leq y$, $\quad y \leq x + 0.5$

examples: $x = 0.499 \Rightarrow 0 \leq y \leq 0.999 \Rightarrow y = 0$
$x = 0.500$ $+0.001 \leq y \leq 1.000 \Rightarrow y = 1$
$x = 0.9$ $0.401 \leq y \leq 1.399 \Rightarrow y = 1$

Absolute value

$$|e_i| = \begin{cases} e_i & \text{if } e_i \ge 0 \\ -e_i & \text{if } e_i < 0 \end{cases}$$

$|e_i| = e_i^+ + e_i^-$ where $e_i^+ = \max\{e_i, 0\} \ge 0$

$e_i = e_i^+ - e_i^-$ $\quad e_i^- = \max\{-e_i, 0\} \ge 0$

LAD regression

$e_i = \hat{y}_i - \sum_{j=1}^{k} w_j x_{ij} - b$, $\quad y_i = \sum_{j=1}^{k} w_j x_{ij} + b + e$

$$\min \sum_{i=1}^{n} |e_i| = \min \sum_{i=1}^{n} e_i^+ + e_i^-$$

subject to $e_i = e_i^+ - e_i^- = \hat{y}_i - \sum_{j=1}^{k} w_j x_{ij} - b$

$e_i^+ \ge 0$

$e_i^- \ge 0$

$w_j \in \mathbb{R}$

$b \in \mathbb{R}$

→ linear optimization model

Application of LAD regression to our DPC problem and OLS regression

$d(p_1, p_2, t) = \beta_{0t} + \beta_{1t} p_1 + \beta_{2t} p_2$

$\hat{d}_t$ observed demand

$d(p_1, p_2, t)$ needs to be rounded first.

$y_t \in \mathbb{N}_0$: $\quad d(p_1, p_2, t) - 0.499 \le y_t \le d(p_1, p_2, t) + 0.5$

$e_t = \hat{d}_t - y_t$

Objective function $\min \sum_{t=1}^{n} e_t^2 = \min \sum_{t=1}^{n} (\hat{d}_t - y_t)^2$

subject to $e_t = \hat{d}_t - y_t$

$d(p_1, p_2, t) - 0.499 \le y_t$

$y_t \le d(p_1, p_2, t) + 0.5$

$(y_t \ge 0 \quad y_t \in \mathbb{N}_0)$